

Day - 30

Data Analysis Project Using AI



A.I. Powered 30 Days Python Micro Course By Satish Dhawale (Microsoft Certified Trainer)

ROLE SETTING (Very Important)

- You are an expert Python software developer and data analyst.
- You have strong experience in:
 - - Pandas for data analysis
 - - Tkinter for GUI applications
 - - Matplotlib for charts
 - - Building corporate-ready desktop automation tools

PROJECT CONTEXT

- I want to build a professional desktop software (GUI-based) for corporate users.
- The software should allow non-technical users to analyze Excel or CSV data
- without writing any Python code.

FUNCTIONAL REQUIREMENTS

Write a complete Python program using Tkinter and Pandas that does the following:



1 File Selection & Reading

- - Allow the user to browse and select a CSV or Excel file.
- - Add a separate "Read" button.
- - When the Read button is clicked:
 - - Read the full dataset
 - - Display:
 - - Total number of rows
 - - Total number of columns
 - - All column headings

2 Dynamic Column Detection

- - Automatically detect:
- - Text columns (object/string dtype)
- - Numeric columns (int/float or numeric-looking text)

3 Report Builder (Interactive Analysis)

- - Provide dropdowns for:
 - 1. Group By Column (only text columns)
 - 2. Aggregation Method (sum, mean, average, max, min, count, median)
 - 3. Value Column (only numeric columns)
- - When user clicks "Preview Report":
 - - Apply GroupBy + selected aggregation
 - - Sort results in descending order
 - - Display the result table inside the GUI (not Excel)

4 Export Report

- - Add an Export button.
- - Allow export format selection:
 - - Excel (.xlsx)
 - - CSV (.csv)
- - Export the report automatically into the same folder
- where the input file is located.

5 Chart Builder (GUI- Based Visualization)

- Provide a dropdown to select chart type:
 - Bar chart
 - Column chart
 - Line chart
 - Pie chart
- When user clicks "Preview Chart":
 - Generate the selected chart using the report data
 - Display the chart inside the GUI window
- Add "Export Chart" option:
 - Export chart as PNG image
 - Save it in the same folder as the input file

6 Usability & Error Handling

- - Show proper error messages if:
 - - File is not selected
 - - Read button is not clicked
 - - Dropdowns are not selected
- - Reset previous outputs when a new report is generated
- - Keep the GUI clean, professional, and easy to use

TECHNICAL CONSTRAINTS

- - Use Tkinter for GUI
- - Use Pandas for data processing
- - Use Matplotlib for charts
- - Write clean, readable, well-commented code
- - Structure the code so it can be converted into a .EXE using PyInstaller
- - Do not require the end user to install Python

DELIVERABLE FORMAT

- - Provide the full Python code in one single block
- - Ensure the program is production-ready
- - Do not omit any required functionality

WHY THIS PROMPT IS PERFECT



Clear role definition



Clear problem statement



Step-by-step requirements



Forces AI to think structurally



Produces enterprise-level code



Teaches real-world AI usage